Datasheet

Black Oak Engineering

precision electronic instruments

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Angel – Tiny Environmental Monitor

Part number BE-ANG1

Version 1.2

Description. The Angel-TEM is a simple, rugged, compact module that continuously monitors and records environmental safety data. Specifically, it senses audio noise level, barometric pressure, humidity, shock

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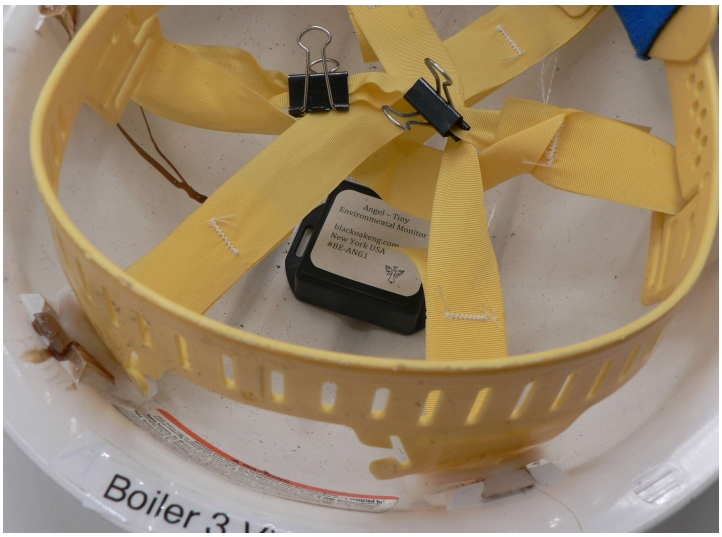
impact, and temperature. It may be used to protect humans, non-human animals, and material assets. Its rechargeable battery allows for 30 day operation under most use cases. It has a programmable audio alarm. It has 256 kB data storage. It has a USB-C (cable included) port for data access and recharging. It has a secure wireless interface for remote operation. It is small enough to fit easily inside a hard hat, helmet, web gear, or other means of securing.

Basic specifications



- ***Audio:*** Sound Level Meter (SLM) measures 35 to 130 dB ±1.4.
- Selectable A- or C-weighting. Frequency response 100 to 10k Hz.
- ***Barometric pressure:*** measures 8.7 to 23.9 PSI (60 to 165 kPa).
- ***Relative Humidity:*** 0.5 to 99 %RH.
- ***Shock:*** 3-axis ±200 g.
- ***Temperature:*** −40 to +85 °C (±0.1).
- Meets or exceeds ISO 26262 requirements.
- Mass: 41.2 grams (1.45 oz).
- Standard 1 m USB-C cable for recharging and data access.
- Diagnostic LED.
- Alarm may be programmed to sound for various environmental conditions, or silenced.
- Rugged ABS construction with heavy-duty mounting flanges suitable for fasteners or straps.
- Wireless: BLE 5.1
    - Range: 75 m (246 ft) typ.
    - Data rate: 2 Mbit/s
    - Security: 128-bit AES
    - Transmit power: 8 dBm max
    - Fully certified for all major international locations. Ask BOE for detailed list.
    - Easily integrated into Safety Compliance Monitoring Systems, or use our free utility.
- Operating temperature range: −30 to +65 °C (−22 to +149 °F).
- Battery recharge temperature range: 10 to +45 °C (50 to +112 °F).
- Compact size: 50 L × 50 W × 20 H mm. (1.97L × 1.97 W × 0.79 H inch), with mounting flanges.
- EAN-13 0729043992928

BOE is here to support you. If you need a modification, contact us. Our Engineering team can often make adjustments to the standard design.

BOE is continuously improving. We also strive to keep one step ahead of procurement shortfalls. We will deliver to you the latest hardware version possible. In some cases specifications will change.

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Approvals & Compliance

- RoHS. REACH.
- California Prop 65. There are no listed toxic materials in the PCBA.
- CE. The system is designed and tested to meet or exceed CE requirements for laboratory, non-safety-critical electronic devices.

Warranty Policy. Any instrument ordered from BOE may be returned for full refund, less shipping costs, within 30 days of delivery, provided that the instrument has not, in the opinion of BOE been damaged or misused. An RMA number is required in all cases. See our *Standard Terms & Conditions – Instruments* for more details.

BOE reserves the right to make changes to these specifications as it deems necessary. All technical information contained herein is as accurate as possible; however BOE shall not be held responsible for any errors or for product use, nor for any infringements upon the rights of others which may result from its use. BOE products are not to be used in life support or safety critical applications.

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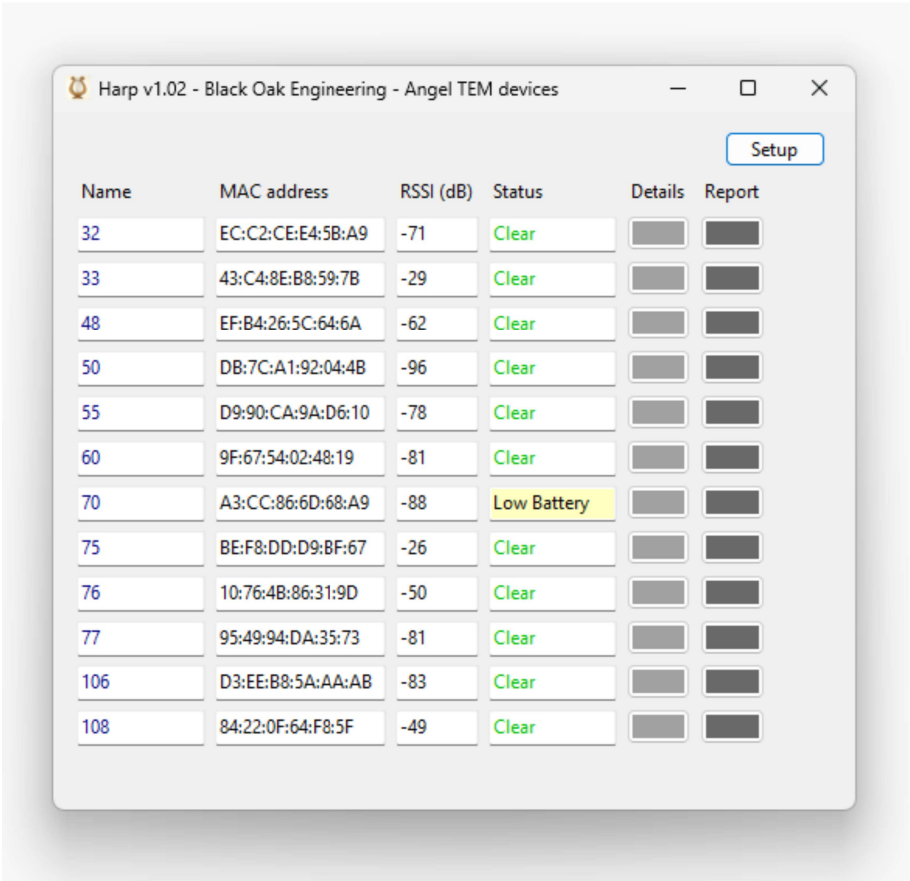

Figure 1: Harp utility for tracking Angel TEMs

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